

Tanasa Dragos

AI SPECIALIST

Pistoia, Italy

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Education

M.S. in Artificial Intelligence

UNIVERSITY OF FLORENCE

Florence, Italy

Sep. 2022 - Apr. 2025

- Developing a strong understanding of the theoretical foundations of AI with a focus on statistics and applications.
- Final GPA: 29.9/30

B.S. in Mechanical Engineer

UNIVERSITY OF FLORENCE

Florence, Italy

Sep. 2019 - Dec. 2022

- Thesis: Machine learning techniques applied to turbulence closure problem in RANS
- Grade: 110/110 with Honors.

Experience

Baker Hughes

AI INTERN

Florence, Italy

Oct. 2024 - today

- Working on the segmentation and mapping of gas leaks in the context of automatic robotic inspection using IR cameras. Leveraging state-of-the-art deep learning models to develop a zero-shot segmentation pipeline and addressing the autoprompting problem in SAM-like models.
- Contributing to the company's fleet manager robotic platform by developing software components that enable real-time streaming and interaction with sensors on the robots using ROS in Python and Docker.
- Performed data analysis and visualization on the results of numerous test campaigns focused on evaluating sensor performance in gas leakage detection and quantification using Python.
- Working in a research-focused, international and multidisciplinary team following the Kanban Agile methodology.

Firenze Race Team (formula student team of UNIFI)

HEAD OF TRAJECTORY AND CONTROLS DEPARTMENT

Calenzano, Italy

Aug. 2023 - Sep. 2024

- Led and managed an eight-member department while developing interdisciplinary competences through cross-departmental collaboration.
- Improved the cone detection pipeline by fine-tuning the YOLOv8 network and integrating it into our autonomous pipeline through ROS. Enhanced vehicle control by creating a dynamics-aware Pure Pursuit algorithm

SOFTWARE ENGINEER

Oct. 2022 - Aug. 2023

- Developed a real time trajectory estimation algorithm from scratch using Delaunay triangulation in Python. Contributed to the control algorithm by implementing the physical model of the car on Symulink for our MPC control. Integrated planning and controls logic in our pipeline using ROS.

Projects

Longformer

DEEP LEARNING, NLP

Jan. 2024

- Implemented Longformer in PyTorch focusing on both the Transformer architecture and an efficient local attention mechanism.
- Reproduce the literature results by performing both pre-training and fine tuning on the most common NLP tasks like classification, question answering and coreference resolution.

Scikit-learn Add-on

MACHINE LEARNING

Jun. 2023

- Designed an add-on library for the scikit-learn framework adhering to the scikit-learn API conventions. Developed several ML algorithms in Python such as: Kernel Regression, Super-Learner, AdaBoost, Gradient Boosting.
- Parallelized code when possible and leveraged optimized libraries such as NumPy, Pandas, and SciPy to achieve maximum efficiency.

Image Processing

COMPUTER VISION

Sept. 2023

- Designed and implemented a highly efficient kernel image processing algorithm in C++ using OpenCV, CUDA and OpenMP.
- Used CUDA and OpenMP to parallelize 2D convolution, achieving a speed-up of 3 compared to the OpenCV implementation
- Used tools like Nsight Compute and Nvidia Visual Profiler to optimize the CUDA implementation.

Other

Martial artist having practiced Karate for ten years and now Brazilian Jiu Jitsu. Passionate reader, I play chess and I'm a speedcuber.